

# Liang Zhang

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## EDUCATION AND TRAINING

### Doctor of Philosophy – Cognitive Neuroscience

Beijing Normal University

September 2019 — July 2024

- Dissertation: Cognitive And Neural Mechanisms Of General Cognitive Abilities: Evidence From Psychometrics And Brain Networks
- Supervised by Prof. Gui Xue

### Research Assistant

Beijing Normal University

August 2017 — August 2019

- Collected and analyzed data from a large-scale longitudinal study of children' s cognitive development
- Used structural equation modeling to explore the structure of children' s cognitive abilities

### Master of Science – Cognitive Neuroscience

Beijing Normal University

September 2014 — July 2017

- Thesis: The Structure and Development Trajectory of Children' s Executive Function
- Graduated as an outstanding graduate of Beijing

### Bachelor of Science – Statistics

Beijing Normal University

September 2009 — July 2013

- Graduated as top 5% of the class
- GPA: 3.7/4.0

## PUBLICATIONS

Sheng, J.#, **Zhang, L.#**, Xue, G\*. (In preparation). Shared and individualized representational transformations support episodic memory formation.

**Zhang, L.**, Feng, J., Liu, C., Hu, H., Zhou, Y., Yang, G., Peng, X., Li, T., Chen, C., & Xue, G. (2024). Improved estimation of general cognitive ability and its neural correlates with a large battery of cognitive tasks. *Cerebral Cortex*, 34(2), bhad510. <https://doi.org/10.1093/cercor/bhad510>

Sheng, J., Wang, S., **Zhang, L.**, Liu, C., Shi, L., Zhou, Y., Hu, H., Chen, C., & Xue, G. (2023). Intersubject similarity in neural representations underlies shared episodic memory content. *Proceedings of the National Academy of Sciences*, 120(35), e2308951120. <https://doi.org/10.1073/pnas.2308951120>

Feng, J., **Zhang, L.**, Chen, C., Sheng, J., Ye, Z., Feng, K., Liu, J., Cai, Y., Zhu, B., Yu, Z., Chen, C., Dong, Q., & Xue, G. (2022). A cognitive neurogenetic approach to uncovering the structure of executive functions. *Nature Communications*, 13(1), 4588. <https://doi.org/10.1038/s41467-022-32383-0>

Sheng, J., **Zhang, L.**, Liu, C., Liu, J., Feng, J., Zhou, Y., Hu, H., & Xue, G. (2022). Higher-dimensional neural representations predict better episodic memory. *Science Advances*, 8(16), eabm3829. <https://doi.org/10.1126/sciadv.abm3829>

Liu, J., Zhang, H., Yu, T., Ren, L., Ni, D., Yang, Q., Lu, B., **Zhang, L.**, Axmacher, N., & Xue, G. (2021). Transformative neural representations support long-term episodic memory. *Science Advances*, 7(41), eabg9715. <https://doi.org/10.1126/sciadv.abg9715>

Feng, J., Chen, C., Cai, Y., Ye, Z., Feng, K., Liu, J., **Zhang, L.**, Yang, Q., Li, A., Sheng, J., Zhu, B., Yu, Z., Chen, C., Dong, Q., & Xue, G. (2020). Partitioning heritability analyses unveil the genetic architecture of human brain multidimensional functional connectivity patterns. *Human Brain Mapping*, 41(12), 3305–3317. <https://doi.org/10.1002/hbm.25018>

#: equal contribution; \*: corresponding author

## CONFERENCE PRESENTATIONS

**Zhang, L.**, Xue, G. The neural substrates of general cognitive ability based on multiple cognitive tasks. Poster presented at the Annual Meeting of the Society for Neuroscience, November 2023, Washington, DC. USA.

## RELEVANT SKILLS

- Programming: R, Python, MATLAB
- Statistical Analysis: Machine Learning, Structural Equation Modeling, Hierarchical Bayesian Modeling.
- Neuroimaging: fMRI, brain network analysis